Privacy VPN

Blockchain-based VPNs: All you need to know

How a new wave of VPNs use the blockchain to ensure decentralization and more

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Table of contents

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With a billion people globally using virtual private networks, it's safe to say VPNs are booming. Until recently, the market has remained unchanged, with little innovation in the field. But now, a wave of next-generation VPNs are hitting the market to solve the problems of traditional VPNs. These are called dVPNs, or decentralized VPNs and they make use of blockchain technology in their infrastructure. But what do blockchains actually do for VPNs?

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What is a blockchain-based VPN?

Traditional, centralized VPNs do not preserve user anonymity. They do mask traffic from your internet service provider. But because the VPN provider encrypts your data and routes it through their own servers, this merely moves trust from your internet service provider to your VPN provider. Everything is owned and overseen by a single entity. As such, there is a central point of failure with traditional VPNs. If your VPN provider is untrustworthy or suffers from poor security, then your browsing habits can be exposed.

The new crop of dVPNs aim to resolve this problem partly by virtue of the fact they are decentralized. They remove this one point of failure and disperse trust across a network of independent 'nodes' instead. A central component here is blockchain.

Blockchains can be implemented anywhere there is an exchange of information that needs to be tamper-free, verifiable, and decentralized, and are already in use by many large organizations. Because they cannot be meddled with, blockchains are useful for anti-fraud record-keeping and provenance, as well as for credentials, identity, and access management.

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processing.

How do VPNs use the blockchain?

In order to create and run decentralized infrastructure, blockchain-based VPNs utilize blockchain technology in a number of ways: to provide incentives for node operators, to decentralize the directory authority, and for authentication and access. Let's take a closer look.

Blockchain-based VPNs: incentivizing nodes for better speed, reliability and uptime

Traditional, centralized VPNs are incentivized to maintain a high network quality by design. Providers rent or own servers and manage these centrally, and try to maintain a good quality of service simply because they need paying customers.

Decentralized VPN providers need to think differently in order to provide reliability, uptime, and good latency levels. One way dVPN providers do this is through rewards and incentives, based on their own blockchain ecosystems. This is important because other decentralized privacy projects, like Tor, are totally run by volunteers. This means that the nodes in the network are only as fast or robust as the volunteers configure them. There's no incentive to provide excellent service other than the goodwill of individual contributors or organizations.

Decentralized VPNs also rely on a network of independent nodes through which to construct traffic paths. However, in contrast, node operators are rewarded for their work. This is usually with the 'token' of that particular project. With Nym, for example, this is the NYM utility token on our Cosmos-based blockchain. With this reward mechanism, node operators are encouraged to optimize their infrastructure for peak performance and with maximum uptime. This reward mechanism is also closely tied to the directory authority mentioned above.

The incentive scheme feeds into the token economics that power NymVPN. These have been carefully crafted to reward high performance and encourage governance within the

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the community member who delegated NYM also receives a share of the rewards from that node.

It is smart contracts hosted on the native Nyx blockchain that governs all these rewards, which also act as the reputation system for nodes on the whole network. If nodes perform poorly, fewer people delegate to them, and they earn fewer rewards.

This mechanism ensures that the Nym network can operate in a fully decentralized manner. Thinking of the bigger picture, these incentives also reward node operators for providing privacy, instead of rewarding them for stealing user data as per the surveillance capitalism model of Web 2.0. Therefore, incentivization is an essential part of decentralized systems, ensuring that they are self-sustaining and do not have to rely on outside support like charitable donations. And it is the blockchain that allows this all to happen.

Blockchain-based VPNs: Decentralized directory authority

The blockchain helps to keep core systems infrastructure decentralized. One essential part of VPNs is the 'directory authority', a system that essentially oversees the health and configuration of servers. With traditional VPNs, this is managed and maintained centrally by the provider. This centralization, however, introduces risk for users - because if the company is compromised, then everything else is too.

Decentralized VPNs do things a little differently.

Decentralized VPNs, like NymVPN, remove this point of failure by using the blockchain for their directory authority. The directory authority of NymVPN is managed on the native Nyx blockchain. The directory authority is decentralized, and the team is working towards making it completely trustless and permissionless. No user data ever touches Nyx.

Blockchains often also use something called 'smart contracts'. This is automated software that can permit transactions or agreements between the blockchain and third

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Rather than having a centralized directory authority, Nym's blockchain, Nyx, uses smart contracts based on the token economics of the project to understand the layout and health of nodes on the network, ensuring decentralization.

Blockchain-based VPNs: Access and payments

Using the blockchain gives dVPNs a decentralized way to manage access.

However, because blockchains are a matter of public record that anyone can look up, providers need to think cleverly in order to anonymize these transactions.

For example, rather than paying for access in regular 'fiat' currencies like dollars, some dVPNs, like the Ethereum-based Orchid and its OXT token, make use of a pseudonymous crypto-based ticketing system. Orchid describes the nanopayments system as akin to a lottery, in order to rent bandwidth or access from individual node operators. However, here, the small number of 'winning tickets' are recorded on the Ethereum blockchain, including timestamps and Ethereum addresses of users and bandwidth providers, meaning this method is not completely anonymous.

Mysterium, another blockchain-based dVPN, uses its own MYST tokens and smart contracts to facilitate transactions between consumers and service providers.

NYM is a utility token, native to the Cosmos-based Nyx blockchain, which enables users to access the internet through NymVPN. To ensure unlinkability between blockchain data and network usage, Nym has also created an anonymous credential cryptographic scheme - a zero-knowledge proof called zk-nyms. This provides unlinkability between payments and NymVPN access credentials that enable clients to communicate through NymVPN, as well as unlinkability between sessions by the same client. This is designed to provide high levels of privacy and prevent client profiling. Watch out for a zk-nyms deep-dive on this blog soon; for now, learn about zk-nyms here.

The best blockchain-based VPN: NymVPN

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With NymVPN, users gain access to a next-generation VPN with two modes – a lightning-fast 2-hop dVPN mode and the 5-hop mixnet mode – offering unparalleled privacy protections and user choice. And the general purpose Layer 1 Nyx blockchain is essential to powering NymVPN, with its smart contracts ensuring infrastructure is decentralized and incentivized.

Nym's Noise Generating Mixnet

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Blockchain-based VPNs: FAQs

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How does blockchain anchoring enhance auditability and transparency in VPN operations?

What consensus mechanisms are used to choose relay nodes in blockchain-based VPNs?

How are micropayments handled to minimize transaction overhead on public blockchains?

What do blockchain-based VPN providers typically do to ensure scalability during high demand?

Are there ways to mitigate censorship or port blocking of blockchain payments in restrictive countries?

About the authors

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These posts are published by the core team behind NymVPN and the Nym mixnet.

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